

MIDDLE PERCY ISLAND, Australia

WMO: 943720

Lat: 21.663S

Lon: 150.271E

Elev: 687

StdP: 14.33

Time Zone: 10.00 (E10)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	57.7	59.0	38.7	35.5	61.4	41.7	39.8	61.6	30.3	61.7	27.5	62.5	16.0	170	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.6	84.0	75.9	83.1	75.5	82.3	75.1	79.1	81.0	78.2	80.3	77.4	79.7	11.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	152.7	80.0	77.6	147.1	79.4	76.7	142.5	78.8	43.4	80.8	42.4	80.6	41.5	79.7	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.1	26.6	24.6	DB	54.8	87.2	2.8	1.7	52.8	88.5	51.2	89.5	49.6	90.4	47.5	91.7	(4)
(5)			WB	46.7	80.2	1.9	1.5	45.3	81.2	44.2	82.1	43.1	82.9	41.7	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.5	78.5	78.3	77.1	73.9	70.1	66.2	65.5	66.2	69.3	72.4	75.2	77.6	(6)	
	DBStd	5.29	1.89	1.89	2.12	1.96	2.77	3.23	3.05	2.72	2.27	1.95	1.98	1.96	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	78	0	0	0	0	2	25	31	19	1	0	0	0	(9)	
	CDD50	8212	882	793	840	718	624	485	481	503	580	695	756	855	(10)	
	CDD65	2817	417	373	375	268	161	61	47	57	132	230	306	390	(11)	
	CDH74	12467	3038	2657	2155	559	83	3	2	6	63	367	1125	2409	(12)	
	CDH80	1014	333	267	139	1	0	0	0	0	3	35	236		(13)	
(14) Wind	WSAvg	13.5	13.8	14.4	15.5	16.0	13.9	13.4	12.9	12.0	11.6	12.7	12.4	13.1	(14)	
(15) Precipitation	PrecAvg	33.1	5.0	5.8	5.3	2.8	2.1	2.0	1.3	1.3	0.8	1.2	1.7	3.9	(15)	
	PrecMax	59.2	13.0	20.5	20.5	8.6	9.3	5.8	7.0	6.6	4.4	4.1	9.1	13.7	(16)	
	PrecMin	14.8	1.1	0.8	0.5	0.3	0.4	0.4	0.1	0.0	0.0	0.1	0.2	0.7	(17)	
	PrecStd	9.8	2.9	4.0	4.9	2.1	1.6	1.3	1.5	1.5	0.9	1.0	2.1	3.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.4	85.2	84.1	80.0	77.6	74.1	73.3	74.8	77.2	79.8	82.6	84.6	(19)	
		MCWB	77.3	76.4	76.9	72.2	71.0	68.5	66.7	67.6	68.1	70.2	73.5	74.7	(20)	
	2%	DB	83.9	83.8	82.7	78.8	76.0	72.5	71.5	72.8	75.6	78.2	80.9	83.3	(21)	
		MCWB	76.3	76.0	75.8	71.7	69.5	67.6	65.7	65.8	67.8	70.3	72.7	74.6	(22)	
	5%	DB	82.9	82.7	81.6	77.8	74.8	71.4	70.3	71.5	74.5	77.2	79.9	82.3	(23)	
		MCWB	75.7	76.0	75.1	71.2	68.6	66.8	65.4	65.0	67.2	69.6	72.0	74.1	(24)	
	10%	DB	81.9	81.8	80.4	76.8	73.7	70.2	69.3	70.1	73.2	76.2	79.0	81.3	(25)	
		MCWB	75.1	75.7	74.5	70.8	67.9	66.0	65.0	64.4	66.5	69.0	71.3	73.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.1	80.0	79.9	75.4	73.2	70.9	70.2	70.6	72.9	74.7	77.1	79.6	(27)	
		MCDB	82.1	82.1	81.2	77.2	75.6	72.0	71.1	71.8	74.4	76.3	79.2	80.9	(28)	
	2%	WB	78.9	78.9	78.4	74.3	71.5	69.1	68.6	68.6	71.0	73.0	75.8	77.9	(29)	
		MCDB	81.4	80.9	80.0	76.3	74.3	71.0	69.5	70.6	72.8	75.4	78.6	79.9	(30)	
	5%	WB	77.9	78.0	77.3	73.4	70.1	68.0	67.6	67.5	69.9	72.0	74.3	76.7	(31)	
		MCDB	80.5	80.3	79.2	75.7	73.1	70.1	68.8	69.7	72.2	74.7	77.7	79.4	(32)	
	10%	WB	77.1	77.2	76.2	72.5	69.1	67.0	66.5	66.2	68.5	71.0	73.1	75.8	(33)	
		MCDB	79.7	79.7	78.5	75.2	72.4	69.2	68.0	68.6	71.2	74.0	76.8	78.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	6.3	5.8	5.5	5.6	5.2	5.4	6.7	7.4	7.4	7.4	7.2	(35)	
		MCDBR	7.5	7.4	7.0	6.2	6.1	6.0	6.0	7.7	8.3	8.2	8.1	8.1	(36)	
	5% WB	MCWBR	3.6	3.8	3.5	3.8	4.3	4.1	4.2	4.6	4.3	4.1	3.8	3.7	(37)	
		MCWBR	6.8	6.9	6.0	5.8	5.8	5.4	5.0	6.7	7.2	7.2	7.5	6.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.389	0.384	0.373	0.347	0.328	0.321	0.309	0.314	0.339	0.364	0.367	0.379	(40)	
		taud	2.567	2.588	2.605	2.666	2.671	2.684	2.700	2.673	2.581	2.535	2.567	2.568	(41)	
	Ebn at Noon		303	301	296	292	285	282	290	298	302	303	308	306	(42)	
		Edh at Noon	34	33	31	27	25	24	24	27	31	35	34	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		2103	1955	1784	1568	1339	1181	1332	1613	1960	2169	2252	2241	(44)	
	RadStd		131	215	157	132	74	112	91	103	122	115	222	218	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page